



Biodiversity responses to agricultural practices in cropland and natural habitats

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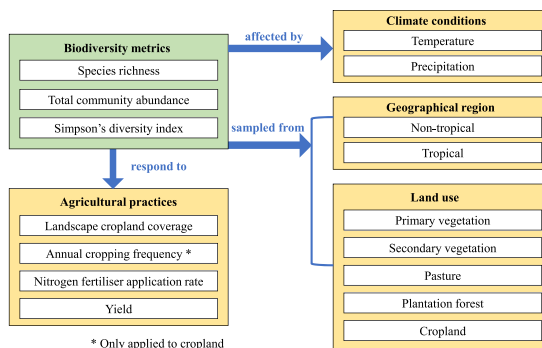
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HIGHLIGHTS

- Whether biodiversity responses to agricultural practices differ among land-use types and geographical regions remain unclear.
- Biodiversity responds to agricultural practices not only in cropland but also in natural habitats.
- Biodiversity in tropical regions is more vulnerable to increased landscape cropland coverage and increased yield.

GRAPHICAL ABSTRACT



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ABSTRACT

Largely driven by agricultural pressures, biodiversity has experienced great changes globally. Exploring biodiversity responses to agricultural practices associated with agricultural intensification can benefit biodiversity conservation in agricultural landscapes. However, the effects of agricultural practices may also extend to natural habitats. Moreover, agricultural impacts may also vary with geographical region. We analyze biodiversity responses to landscape cropland coverage, cropping frequency, fertiliser and yield, among different land-use types

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Landscape cropland coverage
Cropping frequency

and across geographical regions. We find that species richness and total abundance generally respond negatively to increased landscape cropland coverage. Biodiversity reductions in human land-use types (pasture, plantation forest and cropland) were stronger in tropical than non-tropical regions, which was also true for biodiversity reductions with increasing yield in both human and natural land-use types. Our results underline substantial biodiversity responses to agricultural practices not only in cropland but also in natural habitats, highlighting the fact that biodiversity conservation demands a greater focus on optimizing agricultural management at the landscape scale.

1. Introduction

Agricultural land occupied about 36.5 % of the Earth's terrestrial area in 2020 (World Bank, 2022), and intensive agricultural practices have eroded natural habitats more than any other human activity (Foley et al., 2005). The loss, fragmentation and degradation of natural habitats caused by agricultural expansion have been the main driver of biodiversity loss in recent decades (Newbold et al., 2015; Tilman et al., 2001). Agricultural intensification, which typically involves increasing inputs of fertilisers and pesticides, as well as mechanization and intensification of management practices towards shorter rotations and fewer fallow periods, results in additional biodiversity loss, as well as other environmental costs, such as nitrogen loss, greenhouse gas emissions and soil degradation (Gerstner et al., 2014; Tschamtko et al., 2012). Thus, understanding biodiversity responses to both the expansion and intensification of agriculture is a key question for the field of biodiversity conservation.

Many studies overlap maps of agricultural indicators with maps depicting areas of high biodiversity value to identify areas where agricultural activity poses a risk to biodiversity (Bellard et al., 2014; Delzeit et al., 2017; Kehoe et al., 2015; Zabel et al., 2019; Zhao et al., 2022). Nevertheless, those spatial analysis methods mainly reflect the risk of biodiversity loss caused by one integrated agricultural indicator, for instance, cropland expansion potential (Delzeit et al., 2017; Zabel et al., 2019) or agricultural land-use intensity (Kehoe et al., 2015), and do not allow an investigation of biodiversity responses to agricultural practices associated with agricultural intensification. Apart from spatial analysis, large-scale models are widely used to estimate biodiversity responses to land-use pressures (Newbold et al., 2015; Nicholson et al., 2019; Rondinini et al., 2011). However, most large-scale studies examine the influence of agricultural land cover on biodiversity where the interest is on how biodiversity is impacted by agricultural expansion, while there is less study on how agricultural practices associated with agricultural intensification influence biodiversity across large scales (Ortiz et al., 2021). Some research has assessed biodiversity responses to a simple classification of agricultural land-use intensity, which represents a combination of several agricultural practices, such as chemical applications, tillage, mechanization and crop diversity (Gerstner et al., 2014; Millard et al., 2021; Newbold et al., 2018; Newbold et al., 2015; Outhwaite et al., 2022a). However, biodiversity responses to individual agricultural practice remain unclear, although a few broad-scale studies have considered effects of fertiliser and pesticide application rates (Goulson et al., 2015; Millard et al., 2021; Outhwaite et al., 2022b).

Given the prevalence of intensive agriculture accompanied by abundant use of agricultural chemicals, agricultural effects on the environment and biodiversity are profound and extensive, beyond the boundaries of just agricultural lands (Foley et al., 2005). This means that when agricultural practices are intensified, biodiversity may decline not only in agricultural areas, but also in natural habitats adjacent to agricultural areas. Observing these responses requires a large-scale analysis relating how predictors like agrochemicals influence biodiversity within an agroecosystem that encompasses agricultural and natural areas. Therefore, when estimating biodiversity responses to agricultural practices, not only cropland but also other land-use types need to be considered. Otherwise, important facets of biodiversity responses may be omitted.

Biodiversity responses to agricultural practices may also differ between tropical and non-tropical regions. Previous research has demonstrated that non-tropical and tropical biodiversity show different responses to agricultural intensity (Millard et al., 2021; Outhwaite et al., 2022a), which is likely in part because tropical systems have very different environmental conditions (e.g., climate), crop types and associated management practices, and a much shorter history of large-scale intensive agriculture. Moreover, tropical species possess characteristics that are expected to make them disproportionately sensitive to changes in land-use types, such as small range sizes and dietary specialization (Martins and Pereira, 2017; Newbold et al., 2020).

We analyze biodiversity responses to four agricultural practices that are indicative of agricultural intensity or extent: cropping frequency, landscape cropland coverage, nitrogen fertiliser application rate and yield. Cropping frequency depicts the frequency with which crops are cultivated each year, to some extent reflecting agricultural intensity (Ray and Foley, 2013). Increased cropping frequency may require more agrochemical application and more intensive management, thereby affecting biodiversity (Waha et al., 2020). Increasing cropping frequency has been regarded as a promising alternative to cropland expansion for achieving food security, but evaluation of its environmental effects is needed (Mausser et al., 2015; Wu et al., 2018). Cropland where cropping frequency equals to zero has been regarded as fallow lands (Liu et al., 2021), that is, arable lands that have been left without cultivation to control weeds, and recover soil moisture, structure and nutrient levels (Tarjuelo et al., 2020). However, the effects of fallow lands on biodiversity remain unclear, considering that some research has suggested benefits of fallow land for farmland species (Pe'er et al., 2017; Tarjuelo et al., 2020), while other studies have found weak (Dauber et al., 2003) or even negative effects (Koshida and Katayama, 2018) on biodiversity. Increased landscape cropland coverage is associated with reduced coverage of natural habitats, and may also result in habitat fragmentation and increased agrochemical inputs, imposing negative effects on biodiversity (Estrada-Carmona et al., 2022; Fahrig et al., 2011). However, whether the magnitude or even direction of the relationship between landscape cropland coverage and biodiversity varies among land-use types and among geographical regions needs further analysis. Nitrogen fertiliser application rate is used to represent agricultural chemical input. We focus on nitrogen fertiliser, since it is used in most agricultural areas globally, while phosphorous and potassium fertiliser is often applied only on deficient soils (Tenkorang and Lowenberg-DeBoer, 2009). Nitrogen fertiliser is essential to closing yield gaps and meeting global food demands, but at the same time causes widespread nutrient pollution and adverse effects on ecosystems (Tilman et al., 2011). Based on previous research at smaller spatial scales (Devaraju et al., 2022; Isbell et al., 2013; Le Féon et al., 2010), we hypothesize that increased nitrogen fertiliser application rate will have negative effects on biodiversity. Yield is an output-based measure of agricultural intensity, measuring crop production per unit of harvested area. During the past few decades, yields have increased rapidly globally, typically resulting from agricultural intensification, although the correlation between inputs and resulting yields is not always strong (Foley et al., 2011; Ray et al., 2012). Although yield is not expected to have a direct effect on biodiversity, it may capture impactful elements of input intensity for which we lack global estimates. Therefore, we expect indirect negative effects of increased yield on biodiversity. Global

datasets are available for all the agricultural indicators we describe above. However, it is important to acknowledge that there are many potentially important agricultural features for which we lack global estimates. Nevertheless, we expect that combining information on those four indicators will allow new insights into biodiversity responses to agricultural practices.

We use biodiversity measurements from the Projecting Responses of Ecological Diversity In Changing Terrestrial Systems (PREDICTS) database, which is a global collation of studies that sampled species in locations with different land-use type or intensity (Hudson et al., 2014). To account for the hierarchical structure of the PREDICTS database, we apply mixed-effects models to fit responses of sampled species richness, sampled total abundance and the Simpson's diversity index to agricultural practices across land-use types (including natural habitats), and between tropical and non-tropical geographic regions. Specifically, we address three questions. 1) How does biodiversity respond to landscape cropland coverage across land-use types, and to cropping frequency within cropland? 2) How does biodiversity respond to nitrogen fertiliser application rate and yield at the landscape scale? 3) How do these biodiversity responses differ between cropland and natural habitats, and between non-tropical and tropical regions?

2. Materials and methods

2.1. Biodiversity metrics from the PREDICTS database

The PREDICTS database is a collation of individual published studies (or datasets with a published methodology) that compares spatially local biodiversity among different land-use types or land-use intensities (Hudson et al., 2014). The 2016 release of the database, which we used here, contains 3,250,404 measurements, at 26,114 sampling locations worldwide, for 47,044 species. The PREDICTS database has a hierarchical structure, where one published Source may contain one or more Studies, each of which applies a unique sampling protocol. One Study may contain one or more Spatial Blocks, each of which contains several spatially distinct clusters of Sites where biodiversity measurements were sampled.

Taxa sampled within the PREDICTS database comprise vertebrates, invertebrates, plants and fungi. Most individual studies in PREDICTS focus on a particular species group (for example, 70 % of underlying studies sample within a single class), and so the taxonomic focus of biodiversity estimates varies among studies. Resulting differences in biodiversity estimates are accounted for by the hierarchical structure of the models. Measurements may consist of the abundance of an individual taxon (83 % of measurements), occurrence of a taxon (i.e., presence or absence, 17 % of measurements), or summary measures of diversity (e.g., species richness, <1 % of measurements). In this study, we calculated sampled species richness (the number of unique species sampled), Chao-estimated species richness accounting for incomplete sampling (Chao et al., 2005), sampled total abundance (the number of sampled individuals across all detected species), and the Simpson's diversity index (Eq. (1)) (Magurran and McGill, 2011), which collectively capture the main dimensions of local biodiversity (Blowes et al., 2022). It is important to note that we focus on sampled species richness and sampled total abundance, rather than a true estimate of the total species richness or abundance of the ecological community.

$$\text{Simpson's diversity index} = \frac{1}{\sum_{i=1}^S p_i^2} \quad (1)$$

where S represents the number of species sampled; p_i represents the proportion of all sampled individuals that belong to species i . Sampled species richness represents the count of unique sampled taxa, which can be calculated from PREDICTS measurements of occurrence or abundance. Abundance measurements given as the amount in total rather than the amount per unit were assumed to be effort-sensitive and

linearly affected by sampling effort. When sampling efforts differed among sites within one study, we rescaled the raw effort-sensitive abundance measurements, and the rescaling was based on the ratio of each site's sampling effort to the maximum sampling effort across all sites. Sampled total abundance was calculated from summing rescaled abundance measurements of all species in a given site, many of which were not integers owing to this rescaling procedure and some PREDICTS measurements given as the amount per unit. By contrast, we did not rescale species richness for sampling effort, since this procedure would require repeat visits to sites, which most PREDICTS measurements did not conduct. Instead, we calculated Chao-estimated species richness to account for incomplete sampling, which uses properties of sampled abundance to estimate the true number of species within the sampled group present at a site (Chao et al., 2005). We checked the robustness of the responses of sampled species richness by comparing to the responses of Chao-estimated species richness (Supplementary Information S7). We did not present the latter as our main results, because calculation of Chao-estimated species richness requires integer abundance counts, which are available for only around a half of sites in PREDICTS.

2.2. Explanatory variables

Based on the description of the sampled habitat given in the original published data sources, the PREDICTS database classifies site-level land-use type, as follows (Table 1): primary vegetation describes habitats with no record of prior destruction of the natural vegetation; secondary vegetation describes habitats recovering towards their natural state after historical destruction of the vegetation; plantation forest describes areas cultivated for woody crops; cropland describes areas cultivated for herbaceous crops, including for livestock fodder; pasture describes areas grazed regularly or permanently by livestock; urban areas are those used for human residence, buildings, infrastructure and recreation. Urban land use was excluded in this study, as samples were too few for reliable analysis. In addition, based on the sampled sites' latitudes provided by the PREDICTS database, the geographical region was considered as tropical for latitudes between 23.5°S and 23.5°N, and non-tropical outside of these latitudes.

This study incorporated four indicators for agricultural practices from raster datasets (for details, see Table 1), including landscape cropland coverage, cropping frequency, nitrogen fertiliser application rate and crop yields. Expected potential effects of these indicators on biodiversity are shown in Fig. 1. First, based on the thresholds of cropland coverage used in previous studies (FAO, 2022; Liu et al., 2021), landscape cropland coverage were classified into 3 levels based on a 1-km-resolution global map of proportional cropland coverage for 2010 (Cao et al., 2021): low (cropland proportion ≤ 10 %), medium (>10 % and ≤ 60 %) and high (>60 %). Second, cropping frequency was extracted from annual global maps of cropping frequency, which are available for 2001 to 2019 at a spatial resolution of 250 m and have an average overall accuracy of 89 % (Liu et al., 2021). Within the global cropland extent which has excluded abandoned land (Lu et al., 2020), cropping frequency was identified as no cropping (i.e., fallow), single, double and triple cropping, based on the time series of enhanced vegetation index (EVI) from remote-sensed data (Liu et al., 2021). Since triple cropping is scarce, we considered both double and triple cropping as multiple cropping. Third, nitrogen fertiliser application rates were derived from 5-arcmin global mapped estimates for 2000. These estimates were based on national and sub-national statistics on fertiliser application rates and considered 17 major crops (see Supplementary Information S1 for a list), which account for 86 % of global crop calories and around 70 % of all nitrogen and phosphorus application (Mueller et al., 2012; West et al., 2014). Fourth, crop yields were derived from the Spatial Production Allocation Model (SPAM) global data (International Food Policy Research Institute, 2019). Based on a variety of inputs (e.g., crop production statistics, crop suitability, crop prices) (You et al., 2014), the SPAM yield dataset estimated the global distribution of crop

Table 1
Explanatory variables, their data sources, values and spatial resolution.

Variable	Description	Type	Values	Data source	Spatial resolution	Temporal coverage
LU	Land-use type	Categorical	Primary vegetation (PV), Secondary vegetation (SV), Pasture (PA), Plantation forest (PF), Cropland (CR)	(Hudson et al., 2014)	sampled site-level data	Ranging from 2001 to 2013
Region	Geographical region	Categorical	Non-tropical region, Tropical region			
LCC	Landscape cropland coverage	Categorical	Low, Medium, High	(Cao et al., 2021)	1 km	2010
CF	Cropping frequency	Categorical	Fallow, Single, Multiple	(Liu et al., 2021)	250 m	Annually from 2001 to 2013
Fertiliser	Nitrogen fertiliser application rate (kg/hectare)	Continuous	Continuous variables were standardized before fitting models	(Mueller et al., 2012; West et al., 2014)	5 arc min	Circa 2000
Yield	Crop production per harvested area (kg/hectare)	Continuous		(International Food Policy Research Institute, 2019)	5 arc min	2010
Temperature	Mean annual air temperature	Continuous		(Karger et al., 2017)	30 arc sec	The means of 1981–2010
Precipitation	Annual precipitation amount	Continuous				

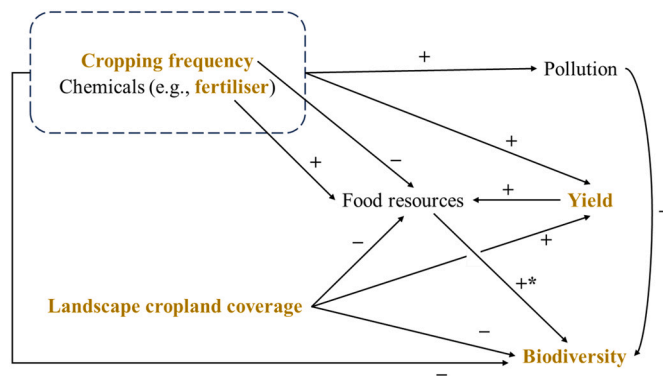


Fig. 1. Diagram showing expected potential effects of agricultural practices on biodiversity. Specific variables considered in this study are shown in brown. *Increased food resources associated with increased crop yields generally promote generalist species that can inhabit disturbed habitats, while narrow-ranged and specialist species may not benefit.

yields for 2010 at a 5-arcmin spatial resolution, for 42 different crops (see Supplementary Information S1 for a list).

Based on geographical coordinates of sampled sites in the PREDICTS database, we extracted the four agricultural indicators for agricultural practices from raster datasets. Of note, yields differ substantially among crops and among regions where specific crops are cultivated. Given that local crop types and crop yields are affected principally by local climate and geographic conditions, we standardized the aggregated yield estimates for 42 crops within combinations of biogeographic realms and biomes (Olson et al., 2001). Besides, the times at which the biodiversity samples in the PREDICTS database were originally made ranged from 1984 to 2013, with 89.83 % sites sampled since 2001. Therefore, based on the sampling year, we matched each PREDICTS measurement that was sampled between 2001 and 2013 with the annual cropping frequency estimates. The other three datasets provide single snapshot rather than annual estimates, and so we could not match them based on the sampling year of the PREDICTS measurements.

Biodiversity and agricultural practices are both strongly impacted by climate (Peters et al., 2019; Rangel et al., 2018). To account for this, we incorporated mean annual temperature and total annual precipitation as explanatory variables in all our models. Climate averages for 1981–2010 were derived from the CHELSA database (Version 2.1) at a 30-arcsecond spatial resolution (Karger et al., 2017).

2.3. Statistical analysis

Given that species richness is an integer count, we built generalized linear mixed-effects models with a Poisson distribution of errors. Since total abundance estimates are often non-integer values, we used linear mixed-effects models, first adding a value of 1 to avoid zero values, and then $\log_e$ -transforming to normalize the residuals. We also fit linear mixed-effects models for $\log_e$ -transformed Simpson's diversity index.

Due to the hierarchical structure of the PREDICTS database, we included random intercepts in our models for study identity and spatial block identity nested within study identity, to capture, respectively, differences in sampling protocols among studies, and differences caused by the spatial arrangement of sites within studies (Newbold et al., 2015). We also included a random intercept for ecoregion (Olson et al., 2001) to account for broad-scale geographical differences in biodiversity. For the species richness models, we also included site identity as an observation-level random intercept to account for the overdispersion present (Rigby et al., 2008). The random-effects structures for each of the models are detailed in Table 2a.

Using an information-theoretic approach (Burnham and Anderson, 2002), we fit a set of candidate mixed-effect models for each biodiversity metric incorporating different combinations of explanatory variables (see Table 2 for details), using the GLMER function from the StatisticalModels package in R (Newbold, 2023). To improve model convergence, we standardized all continuous explanatory variables, so that the mean and the standard deviation of values were zero and one, respectively (Zuur et al., 2009). Climate is known to have non-linear effects on biodiversity (Tukiainen et al., 2017), and so we fit quadratic relationships of each of the biodiversity metrics with mean annual temperature and total annual precipitation. Preliminary models including just fertiliser application rate and climate as fixed effects suggested stronger support for linear rather than quadratic effects of fertilisers (i.e., models with the linear form had a lower Akaike Information Criterion (AIC)), and so the final set of models considered only a linear relationship. Similarly, the linear rather than quadratic effects of yield had stronger support, and so the linear form of yield was used in all candidate models.

We developed 5 different classifications of sites, based on different combinations of land-use type (LU), landscape cropland coverage (LCC) and cropping frequency (CF) (Table 2b). For the most complex combined variable (LU.LCC.CF), the combinations of "low landscape cropland coverage and single cropping" and "low landscape cropland coverage and multiple cropping", which had insufficient representation (<50 sites) in the dataset, were excluded. We considered each of these 5 categorical classifications within each of 12 combinations of the other explanatory variables (Table 2c), giving a total of 60 candidate models.

Table 2

Random-effects and fixed-effects structures of all candidate models. The term “Class” in formulae 1–12 in (c) can be any one of the 5 categorical classifications shown in (b), resulting in a total of 60 combinations.

a. Random-effects structure		
Biodiversity metric	Random-effects structure	Model error structure
Species richness	(1 Study) + (1 Spatial Block) + (1 Site) + (1 Ecoregion)	Poisson
Total abundance	(1 Study) + (1 Spatial Block) + (1 Ecoregion)	Gaussian
Simpson's diversity index	(1 Study) + (1 Spatial Block) + (1 Ecoregion)	Gaussian

b. Different classifications of sites	
Name	Classification
Class1	LU
Class2	LCC
Class3	LU.CF
Class4	LU.LCC
Class5	LU.LCC.CF

c. Fixed-effects structure	
Formula Name	Fixed-effects structure
Formula.Null	poly(Temperature,2) + poly(Precipitation,2)
Formula.Region	Region + poly(Temperature,2) + poly(Precipitation,2)
Formula1	Class + poly(Temperature,2) + poly(Precipitation,2)
Formula2	Class*Region + poly(Temperature,2) + poly(Precipitation,2)
Formula3	Class + Fertiliser + poly(Temperature,2) + poly(Precipitation,2)
Formula4	Class + Yield + poly(Temperature,2) + poly(Precipitation,2)
Formula5	Class*Region + Fertiliser + poly(Temperature,2) + poly(Precipitation,2)
Formula6	Class*Region + Yield + poly(Temperature,2) + poly(Precipitation,2)
Formula7	Class*Region + Fertiliser + Yield + Fertiliser:Yield + poly(Temperature,2) + poly(Precipitation,2)
Formula8	Class*Region + Fertiliser + Region:Fertiliser + poly(Temperature,2) + poly(Precipitation,2)
Formula9	Class*Region + Yield + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)
Formula10	Class*Region + Fertiliser + Yield + Fertiliser:Yield + Region:Fertiliser + poly(Temperature,2) + poly(Precipitation,2)
Formula11	Class*Region + Fertiliser + Yield + Fertiliser:Yield + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)
Formula12	Class*Region + Fertiliser + Yield + Fertiliser:Yield + Region:Fertiliser + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)

In addition to these models, there was also a null model that included only the climate variables, and a model that considered only geographical region (tropical/non-tropical) and climate variables, giving a final total of 62 candidate models. Given that measures of yield and fertiliser are at the landscape scale, we assumed their interactions with land-use type were hard to be detected, which were not considered. Besides, we were primarily interested in the interactions between geographical region and the other variables, and incorporating too many interactions can result in overly complex models.

For each biodiversity metric, we calculated AIC, the difference in AIC value compared with the model with the lowest AIC (ΔAIC) and AIC weight for each candidate model. AIC weight represents the relatively probability of each model being the most appropriate among the candidate models considered, and is calculated following Eq. (2) (Burnham and Anderson, 2002; Zuur et al., 2009).

$$AIC\ weight_i = \frac{e^{-\frac{1}{2} \times \Delta AIC_i}}{\sum_i e^{-\frac{1}{2} \times \Delta AIC_i}} \quad (2)$$

where I is the total set of candidate models, ΔAIC_i is the difference between the AIC for model_{*i*} and the model with the lowest AIC.

We ranked candidate models in the descending order of AIC weights, and calculated the cumulative sum of their AIC weights until the cumulative AIC weight exceeded 0.95. We identified the 95 % confidence set of models containing the best-fitting models with a cumulative AIC weight > 0.95. The relative importance of a variable was calculated by summing the AIC weights of candidate models which incorporated that variable. Following the method proposed by Nakagawa and Schielzeth (2013), we calculated conditional pseudo R^2 and marginal pseudo R^2 for each of the models in the 95 % confidence set. The former represents the variation in the response variable explained by both the fixed and random effects, and the latter represents the variation explained by the fixed effects only.

To visualize biodiversity responses and associated uncertainty, we generated 1000 predictions of biodiversity as a percentage compared to the baseline (primary vegetation in low-cropland landscapes). For each prediction, one model was drawn at random from among models in the 95 % confidence set, weighted by AIC weight, and final predictions were drawn at random from a multivariate normal distribution based on the mean coefficient estimates and the variance-covariance matrix of the model. We set primary vegetation with low landscape cropland coverage as the baseline. For scenarios which contained different combinations of land-use types, landscape cropland coverage and cropping frequency, their predicted values of each biodiversity metric were compared with the predicted values for the baseline scenario, and thus were expressed as a percentage compared to this baseline. Meanwhile, the predicted responses of biodiversity were presented as changes in biodiversity with agricultural practices being intensified.

2.4. Model checks

First, for candidate model with the highest AIC weight for each of the biodiversity metrics (Supplementary Information S2), we conducted a set of graphical analyses to check model assumptions (Supplementary Information S3): 1) Q-Q plots and 2) plots of fitted against observed values were used to check the linearity assumption. 3) To test for residual spatial autocorrelation, we conducted a Moran's I test on the residuals of each Study, and assessed the proportion of studies where the residuals showed significant spatial autocorrelation (a significant result is expected by chance for 5 % of studies). 4) For abundance models and Simpson's diversity index models, we plotted fitted values versus residuals to check for homogeneity of variance.

Second, we checked multicollinearity among explanatory variables in the models with the most complex fixed-effects structure for each biodiversity metric (Supplementary Information S2), by calculating variance inflation factors using the `vif` function from the `car` package version 3.0-10 in R (Fox and Weisberg, 2019). We extracted generalized variance inflation factors (GVIF), adjusted for degrees of freedom (Df) as $GVIF^{(1/(2 \times Df))}$ (Fox and Monette, 1992). Given that $GVIF^{(1/(2 \times Df))}$ values for candidate models with the most complex fixed-effects structures were all <2, indicating a low degree of collinearity, we did not exclude any variables from the models on this basis (Supplementary Information S4).

In our main models, we fitted $\log_e$ -transformed total abundance and $\log_e$ -transformed Simpson's diversity index as response variables in linear mixed-effects models. However, log transformation of count data can result in poor model performance (O'Hara and Kotze, 2010). As a robustness check, we fit zero-inflated negative binomial (ZINB) models on species richness, untransformed total abundance, and untransformed Simpson's diversity index (Supplementary Information S5). These ZINB

models had the same fixed-effects and random-effects structure as the candidate models with the highest AIC weights for each of the corresponding original biodiversity metrics, generated similar results (Supplementary Figs. S7–S9), and did not show obvious improvements in conforming to model assumptions (Supplementary Figs. S10–S12).

Fourth, to test the robustness of our candidate models to violation of the assumptions of standard parametric statistical tests, we fit a set of models using a Bayesian Markov Chain Monte Carlo (MCMC) approach, using the MCMCglmm package version 2.34 in R (Hadfield, 2010). These MCMC models had the same fixed-effects and random-effects structures as the original candidate models with the highest AIC weight for each biodiversity metric. We set the MCMC iterations as 60,000, the thinning interval as 10, and the burn-in period as 3000 iterations, and adopted the default weakly informative priors of MCMCglmm. We visually inspected the trace plots of MCMCglmm models to check model convergence. Coefficients of fixed effects were very similar between the candidate models and the MCMC models (Supplementary Information S6).

3. Results

3.1. The 95 % confidence set of candidate models for biodiversity metrics

For each biodiversity metric, we identified its 95 % confidence set of models (see the full set in Table S1) from all the candidate models showed in Table 2, based on AIC weights, and used this set of models to predict biodiversity responses. In the 95 % confidence set of models, three candidate models with the highest AIC weights are showed in Table 3. The numbers of models in the 95 % confidence set for species richness (15), total abundance (9) and Simpson's diversity index (16) indicated a good degree of convergence on a set of likely explanatory variables. For each biodiversity metric, conditional and marginal pseudo R^2 were almost identical among models in the 95 % confidence set, which were approximately 0.62 and 0.014 for species richness, 0.89 and 0.018 for total abundance, and 0.78 and 0.036 for Simpson diversity index.

Based on relative variable importance among all candidate models (Table 3d), landscape cropland cover, cropping frequency, fertiliser and yield had relatively important effects on species richness; landscape cropland cover, fertiliser and yield had relatively important effects on total abundance; and cropping frequency and yield had relatively important effects on Simpson's diversity index. For continuous variables, we also calculated the model-averaged estimates for their effect sizes (Burnham and Anderson, 2002). For species richness, total abundance and Simpson's diversity index, the estimated effect sizes for fertiliser were -0.38 , 0.41 and -0.23 , respectively, while the estimated effect sizes for yield were -0.46 , -0.61 , -1.74 , respectively. Since our candidate models included different combinations of land-use type, landscape cropland cover and cropping frequency (Table 2b), the classifications of those combined variables cannot be matched across candidate models incorporating different combined variables. Therefore, model-averaged estimates for effect sizes were not available for landscape cropland coverage or cropping frequency. As a reference, for the candidate models with the highest AIC weights, we provided the estimated effect sizes for all explanatory variables (see Tables S9, S10 and S11).

3.2. Biodiversity responses across land-use types to landscape cropland coverage and cropping frequency

The effects of anthropogenic land-use types on biodiversity were much stronger in tropical regions than in non-tropical regions (Fig. 2). In pasture, plantation forest and cropland, relative to the baseline (primary vegetation with low landscape cropland coverage), the largest reductions in the tropical regions were greater than those in the non-tropical regions for all biodiversity metrics. For the example of pasture

Table 3

Three candidate models with the highest AIC weights, and relative importance of explanatory variables. ΔAIC is the difference in AIC between each model and the model with the lowest AIC. AIC weight represents the relatively probability of each model being the most appropriate among the candidate models. N_{para} represents the number of parameters in the model. LU represents land-use types. LCC represents landscape cropland coverage. CF represents cropping frequency.

a. Species richness				
Model	ΔAIC	AIC weights	N_{para}	
LU.LCC*Region + poly(Temperature,2) + poly(Precipitation,2)	0	0.14	34	
LU.LCC*Region + Yield + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)	0.07	0.14	36	
LU.LCC*Region + Yield + poly(Temperature,2) + poly(Precipitation,2)	0.34	0.12	35	
b. Total abundance				
Model	ΔAIC	AIC weights	N_{para}	
LU.LCC*Region + poly(Temperature,2) + poly(Precipitation,2)	0	0.25	34	
LU.LCC*Region + Fertiliser + Yield + Fertiliser:Yield + Region:Fertiliser + poly(Temperature,2) + poly(Precipitation,2)	1.07	0.15	38	
LU.LCC*Region + Yield + poly(Temperature,2) + poly(Precipitation,2)	1.49	0.12	35	
c. Simpson's diversity index				
Model	ΔAIC	AIC weights	N_{para}	
LU.CF*Region + Yield + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)	0	0.27	20	
LU*Region + Yield + Region:Yield + poly(Temperature,2) + poly(Precipitation,2)	0.64	0.19	16	
LU.CF*Region + Yield + poly(Temperature,2) + poly(Precipitation,2)	1.61	0.12	19	
d. Relative importance of explanatory variables				
Biodiversity metric	LCC	CF	Fertiliser	Yield
Species richness	1.00	0.37	0.37	0.56
Total abundance	1.00	0.03	0.56	0.61
Simpson's diversity index	0.07	0.54	0.23	0.92

in a high-cropland landscape, non-tropical species richness was lower than the baseline by 18 % (Fig. 2b) while tropical species richness was lower than the baseline by 40 % (Fig. 2c).

Between low and high landscape cropland coverage, primary vegetation generally experienced increases in species richness, (non-tropical by 12 %; tropical by 1 %) and total sampled abundance (non-tropical by 9 %; tropical by 20 %). In secondary vegetation, species richness showed no clear relationship with landscape cropland coverage, while total abundance declined in the non-tropical region by 9 % but increased in the tropical region by 10 %. For cropland, species richness and total abundance increased with cropland coverage in the non-tropical region by 6 % and 28 %, but declined in the tropical region by 13 % and 31 %. For pasture, both species richness (non-tropical by 8 %; tropical by 9 %) and total sampled abundance (non-tropical by 21 %; tropical by 15 %) decreased with landscape cropland coverage. In pasture with low landscape cropland coverage and in plantation forest with medium landscape cropland coverage, non-tropical total abundance was obviously greater than the baseline. For Simpson's diversity index, landscape cropland coverage was included in 2 candidate models (Table 3c), but had low relative variable importance (Table 3d).

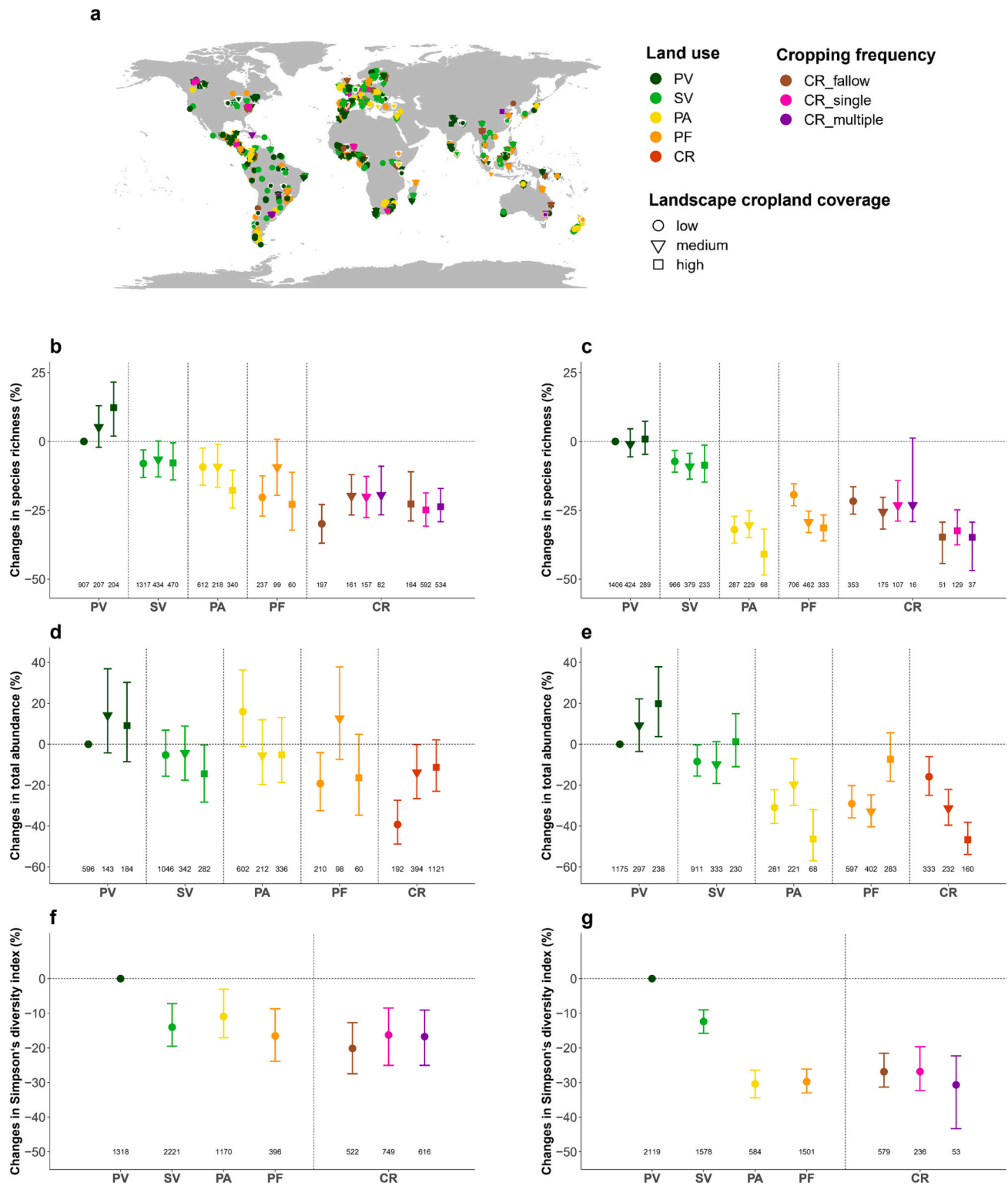


Fig. 2. Responses of biodiversity to land-use types, landscape cropland coverage and cropping frequency. (a) Locations of sites contributing data to the analyses shown in this figure. Responses of species richness (b, c), total abundance (d, e) and Simpson's diversity index (f, g) in non-tropical (b, d, f) and tropical regions (c, e, g). All sites shown in (a) have valid records for species richness, but some sites (bordered in white) lack records for total abundance or Simpson's diversity index. The legend includes all colours and shapes used among panels: in (a), (b) and (c), all land-use types are classified by landscape cropland coverage while cropland is further classified by cropping frequency; in (d) and (e), all land-use types, including cropland, are only classified by landscape cropland coverage; in (f) and (g), land-use types are not classified by landscape cropland coverage, but cropland is classified by cropping frequency. In all panels, there are 5 land-use types: primary vegetation (PV), secondary vegetation (SV), pasture (PA), plantation forest (PF) and cropland (CR). Primary vegetation in a low-cropland landscape is taken as the baseline, and changes in biodiversity are expressed as a percentage compared to this baseline. The estimates of biodiversity responses were generated as the median values of 1000 predictions, while the 95 % confidence intervals were determined by 2.5th and 97.5th percentiles of 1000 predictions. Error bars represent 95 % confidence intervals.

Fallow lands may support ample food resources and habitats, and thereby can harbour high levels of biodiversity (Tarjuelo et al., 2020). However, we found that in fallow cropland, non-tropical species richness was between 20 % and 30 % lower than the baseline (primary vegetation in a low-cropland landscape), and tropical species richness was between 22 % and 35 % lower, depending on landscape cropland coverage. Simpson's diversity index for fallow cropland was 20 % lower than the baseline in the non-tropical region and 27 % lower in the tropical region. These reductions were close or even slightly greater compared to single-cropped and multiple-cropped systems.

3.3. Biodiversity responses to fertiliser and yield

We analyzed differences in biodiversity responses to fertiliser and yield between tropical and non-tropical regions. The effects of nitrogen fertiliser application rate on both species richness and total abundance were weak (Fig. 3). Owing to the low relative importance of fertiliser among candidate models for Simpson's diversity index (Table 3d), we only analyzed the responses of species richness and total abundance to yield.

As yield increased across the sampled range of values, species richness declined by 0.45 % in the non-tropical region and by 2.30 % in the tropical region (Fig. 4). Simpson's diversity index showed a stronger declining trend than the other two biodiversity metrics, decreasing by 2.76 % in the non-tropical region and 10.70 % in the tropical region with increases in crop yield across the sampled range, although uncertainty in the fitted relationships remained high. Total abundance showed a negligible response to crop yield.

4. Discussion

Our results have demonstrated that in both human land-use types (pasture, plantation forest and cropland) and natural habitats (primary vegetation and secondary vegetation), agricultural practices substantially affect local biodiversity. Biodiversity generally decreased in human land-use types, or when landscape cropland coverage increased. All three biodiversity metrics responded more strongly to anthropogenic land-use types and landscape cropland coverage in tropical compared to non-tropical regions, but showed relatively little variation with respect to landscape cropland coverage in primary and secondary vegetation. Contrary to some previous studies (Hertzog et al., 2023; Sanz-Prez et al.,

2019), we found that fallow lands did not maintain higher levels of biodiversity than single-cropped or multiple-cropped systems. Unexpectedly, the effects of increased nitrogen fertiliser application on both species richness and total community abundance were weak and mixed. Consistent with previous research (Jones et al., 2023; Zewdie et al., 2022), increased yield negatively affects biodiversity and more strongly so in tropical regions.

Our results show that landscape cropland coverage significantly affects species richness and total abundance. The negative effects of increased landscape cropland coverage are possibly induced by the simplified landscape composition and configuration (e.g., a smaller amount of remaining natural habitat) (Estrada-Carmona et al., 2022), as well as the greater overall levels of profound and extensive pollution that comes with more intensive field management (Tilman et al., 2001). By comparison, in primary vegetation, species richness and total abundance were estimated to be higher in medium-cropland landscape or high-cropland landscape than in low-cropland landscape. However, between different cropland coverage in primary vegetation, there is substantial overlap between error bars of predicted biodiversity responses, and thus those responses are not significantly different.

Consistent with previous research (Newbold et al., 2020; Outhwaite et al., 2022a), our finding confirms a disproportionate sensitivity of tropical biodiversity to land-use change. Moreover, we have further demonstrated that the disproportionate sensitivity of tropical biodiversity occurs in response also to increased landscape cropland coverage, as well as to land-use type. This sensitivity may be caused by tropical species typically being dietary and habitat specialists, and often having narrower range sizes than non-tropical species (Forister et al., 2015; Newbold et al., 2018; Slatyer et al., 2013). Alternatively, the elimination of sensitive species has occurred earlier and to a greater extent in temperate than in tropical regions, since the former has already undergone a longer history of human-induced land-use change (Balmford, 1996), thus typically leaving a more resistant set of species in non-tropical regions that respond less to contemporary land-use changes. Of note, we acknowledge that this is not always the case and the difference in land-use history is just one of the potential reasons, considering land-use history also vary with biogeographic realms. For instance, differences in land-use history are larger between Nearctic and Palearctic regions than between Nearctic and Neotropical regions (Klein Goldewijk et al., 2011). Another potential reason is that compared to non-tropical regions, agricultural development in tropical developing

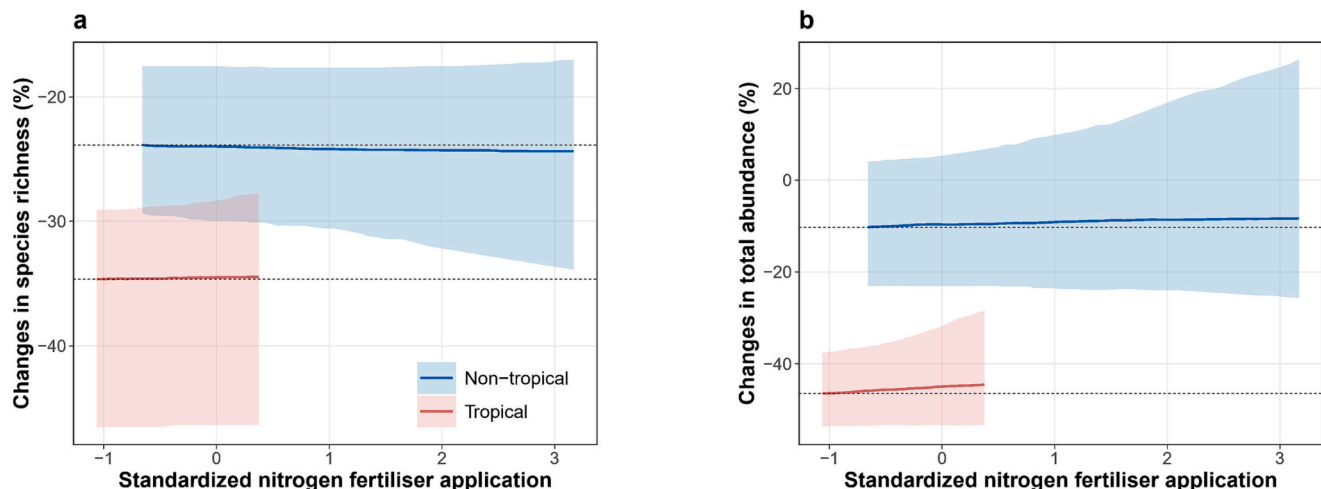


Fig. 3. Responses of biodiversity to standardized nitrogen fertiliser application rate in the surrounding landscape. Changes in (a) species richness and (b) total abundance were expressed as a percentage compared to the baseline, which was primary vegetation with low landscape cropland coverage and with minimal nitrogen fertiliser application rate. In all panels, relationships are plotted for standardized nitrogen fertiliser application rates between the 2.5th and 97.5th percentiles of values for sampled sites. Lines represent the median of 1000 fitted biodiversity estimates; shaded areas represent the 95 % confidence intervals of the slope coefficients across the 1000 estimates. The other continuous explanatory variables were set as the median values across sampled sites.

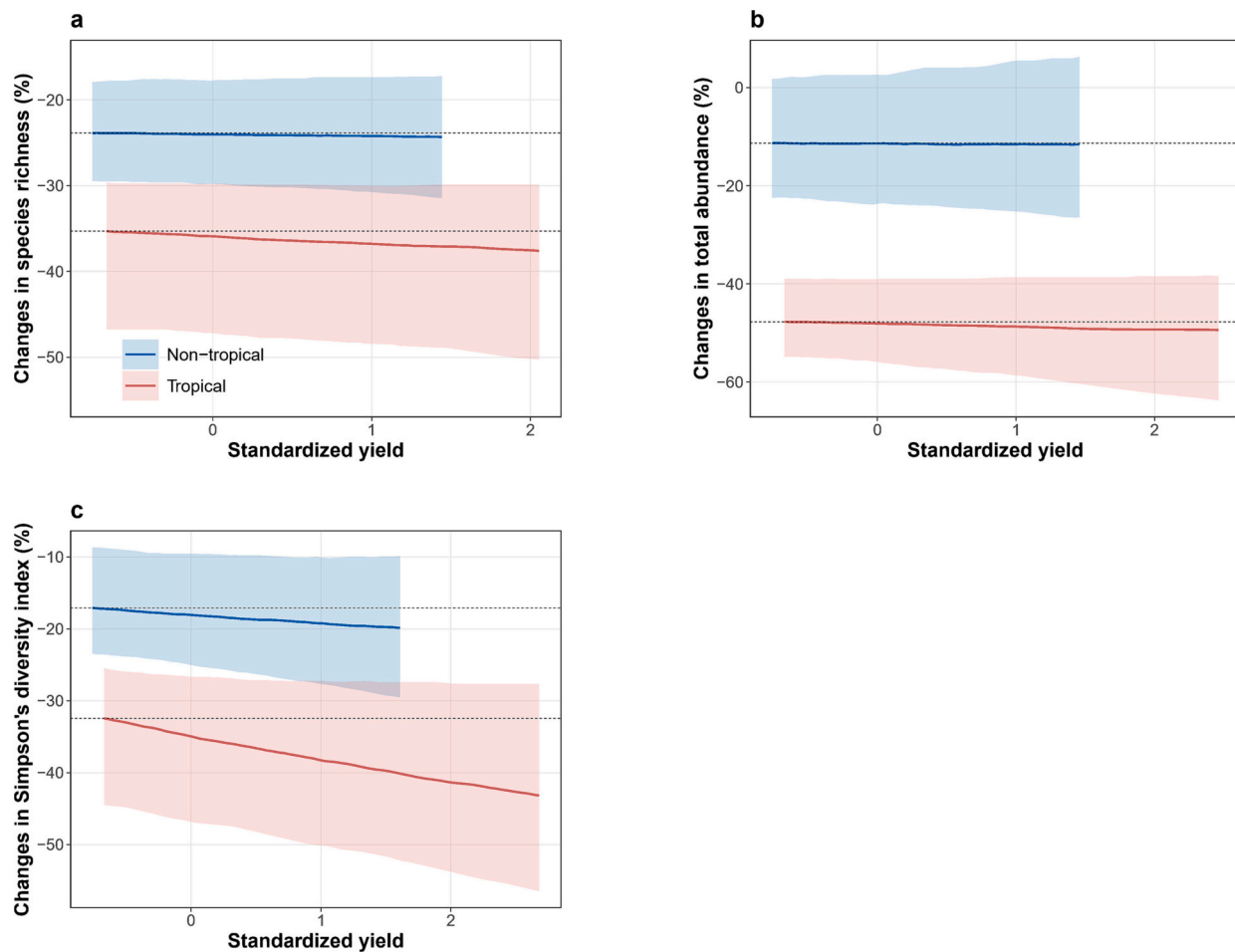


Fig. 4. Responses of biodiversity to standardized agricultural yield in the surrounding landscape. Changes in (a) species richness, (b) total abundance and (c) Simpson's diversity index were expressed as a percentage compared to the baseline, which was primary vegetation with low landscape cropland coverage and with minimal yield. In all panels, relationships are plotted for standardized agricultural yield between the 2.5th and 97.5th percentiles of values for sampled sites. Lines represent the median of 1000 fitted biodiversity estimates; shaded areas represent the 95 % confidence intervals of the slope coefficients across the 1000 estimates. The other continuous explanatory variables were set as the median values across sampled sites.

countries requires improving transportation and irrigation infrastructure which grows rapidly and causes a substantial increase in the agricultural footprint (Laurance et al., 2014). Furthermore, humid tropical regions likely demand more application of fertilisers, lime and pesticides to maintain agricultural production (Laurance et al., 2014), due to the infertility and acidity of soils and high risk of pest attack. Therefore, compared to temperate agriculture, these features of tropical agriculture possibly result in more agrochemical inputs and more intensive management when landscape cropland coverage increases, thereby exerting a stronger effect on biodiversity, besides the generally higher sensitivity to disturbance of tropical species.

Fallow management seems to fail to promote local species richness and Simpson's diversity in both tropical and non-tropical regions, which may be for a few possible reasons. First, herbicides can be heavily applied on fallow lands for weed control (Tarjuelo et al., 2020) or may drift from nearby fields (Qi et al., 2020). Second, as fallow lands will ultimately be ploughed for cropping, they generally include semi-natural vegetation, but cannot achieve the full recovery of natural vegetation (Tarjuelo et al., 2020; Traba and Morales, 2019). Consequently, the habitats and food resources provided by fallow lands may still be limited. Third, although previous research has demonstrated benefits from fallow cropland to biodiversity, these benefits are limited to certain species and are affected by the structure of fallow lands (e.g., the establishment method, seed mixture and local soil conditions) and local management (Kuussaari et al., 2011; Traba and Morales, 2019).

Contrary to our expectation, increased nitrogen fertiliser application showed weak and mixed effects on species richness and total abundance, which can to some extent be explained by its complex effects on the environment. Increased nitrogen fertiliser application may cause pollution or eutrophication (Isbell et al., 2013; Sanchez-Bayo and Wyckhuys, 2019), but can also promote plant productivity and result in more food resources for higher trophic levels (Tscharntke et al., 2005). Besides, the fertiliser dataset we used had a relatively low spatial resolution of 5 arc minutes, which likely challenges the detection of effects on local biodiversity. By contrast, increased yield has negative effects on all three biodiversity metrics, especially tropical Simpson's diversity index, although these effects were generally very weak. The negative relationship between yield and biodiversity is expected to be through indirect effects, since high yield is typically achieved by high levels of inputs and intensive management (Foley et al., 2011). Our results are consistent with previous research which has detected a negative relationship between yield and biodiversity (Jones et al., 2023; Zewdie et al., 2022), although some studies have demonstrated the possibility of maintaining biodiversity without compromising agricultural yield (Clough et al., 2011; Tamburini et al., 2020).

Our analyses have some limitations. First, our analyses are based on spatial comparisons of biodiversity, which neglects potential time-lagged responses. Specifically, we used a space-for-time approach, which has been widely used in ecological modeling, to infer changes of biodiversity over time from contemporary spatial patterns (Attinello

et al., 2023; Blois et al., 2013). Although analyzing biodiversity changes through time can help to understand time-lagged effects, the lack of long time series of global biodiversity and agricultural practices makes the use of spatial analyses necessary (Millard et al., 2021; Purvis et al., 2018). Second, owing to limited data availability, we had to use datasets describing agricultural practices at very different spatial resolutions (Table 1), which may affect our results. Specifically, we acknowledge that biodiversity responses to yield and fertiliser were relatively weak in general. However, our results could still show that biodiversity responses to fertiliser application rates were mixed, while biodiversity responses to yield were stronger, especially in tropical regions and for Simpson's diversity index. Third, the fixed effects in our models have limited explanatory power, which is common in previous studies using the PREDICTS database (Millard et al., 2021; Newbold et al., 2015), and other analogous heterogeneous biodiversity databases (Spooner et al., 2018; Williams et al., 2022). Even so, our models are still effective in assessing the direction and magnitude of local biodiversity responses. Fourth, future research could incorporate more agricultural indicators, such as irrigation, crop diversity, and mechanization levels (Laurance et al., 2014; Tscharnke et al., 2005). Fifth, this study has considered all taxa equally and analyzed them all together, and thereby may underestimate the influence of agricultural practices on specific groups, since different taxa show different sensitivity to agriculture (Gonthier et al., 2014; Pocock and Jennings, 2008). Future research should further focus on responses of different taxa to agricultural practices.

5. Conclusions

Overall, we have demonstrated that biodiversity responds to agricultural practices not only within cropland but also in natural habitats, which has implications for meeting the Kunming-Montreal global biodiversity targets (Secretariat of the Convention on Biological Diversity, 2022). We further show that biodiversity within anthropogenic land-use types is more vulnerable to intensified agricultural practices in tropical than non-tropical regions, and that fallow lands do not maintain higher levels of biodiversity than single-cropped or multiple-cropped systems. We highlight that apart from maintaining natural habitats and establishing protected areas, local biodiversity conservation also depends on appropriate restrictions being placed on nearby agricultural development, including but not limited to cropland expansion and management intensification. Optimization of current agricultural management and practices is an urgent necessity in tropical regions, since these regions generally harbour abundant but sensitive species, which show strong negative responses to intensified agricultural practices.

CRedit authorship contribution statement

Jianqiao Zhao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Conceptualization. **Le Yu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Tim Newbold:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Xiaoli Shen:** Writing – review & editing, Visualization. **Xiaoxuan Liu:** Writing – review & editing, Resources. **Fangyuan Hua:** Writing – review & editing, Methodology. **Kasturi Kanniah:** Writing – review & editing, Visualization. **Keping Ma:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data we used is from public datasets and could be downloaded

through the corresponding reference.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2024.171296>.

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